

Under the Hood: High-Speed Impact Kinematics

Simulation Engine V2.0 Physics Architecture

Engineering Club

1 Introduction: The Mechanics of a Crash

In motorsport, the survival of a driver during an impact relies on the careful management of kinetic energy ($KE = \frac{1}{2}mv^2$). If an F1 car hits a wall and stops instantly, the deceleration forces (measured in Gs) would be fatal.

To prevent this, the carbon-fibre nose cone acts as a sacrificial “crumple zone.” It crushes over a specific distance (d), doing *Work* ($W = F \times d$) to bleed off the energy. A softer nose cone means a longer crush distance, which stretches the impact over more time, drastically lowering the peak force the driver feels.

2 Elasto-Plastic Causality: The Piecewise Model

Our physics engine models the F1 nose cone as a non-linear spring. Initially, the carbon-fibre compresses elastically (stiffness k_e). However, once the force exceeds the material’s yield point (x_{yield}), it breaks and enters a plastic deformation phase (stiffness k_p).

The restoring force $F(x)$ pushing back against the car based on the crush distance x is:

$$F(x) = \begin{cases} k_e x & \text{if } x \leq x_{\text{yield}} \\ k_e x_{\text{yield}} + k_p (x - x_{\text{yield}}) & \text{if } x > x_{\text{yield}} \end{cases} \quad (1)$$

By Newton’s Second Law, the acceleration of the car at any point during the crash is:

$$\frac{d^2x}{dt^2} = -\frac{F(x)}{m} \quad (2)$$

This is a second-order Ordinary Differential Equation (ODE). Because of the extreme stiffness of carbon fibre ($k_e \approx 500,000 \text{ N/m}$), the forces spike incredibly fast.

3 Rigorous Integration: The RK4 Solver

If we attempted to solve this high-stiffness collision using the basic Euler method, the simulation would “explode”—the massive forces would overshoot in a single 16-millisecond frame, causing the math to spiral to infinity.

To guarantee absolute stability, our Engine decouples the second-order ODE into a system of two first-order ODEs:

$$\frac{dx}{dt} = v, \quad \frac{dv}{dt} = a(x, v) = -\frac{F(x)}{m} \quad (3)$$

We then feed this system into a **Runge-Kutta 4th Order (RK4)** integrator. RK4 is an industrial-grade numerical algorithm used in Finite Element Analysis (FEA) by actual F1 engineers. Instead of blindly stepping forward, RK4 evaluates four intermediate gradients (slopes) per timestep ($h = \Delta t$):

$$\begin{aligned}
 k_{1,x} &= v_n, & k_{1,v} &= a(x_n, v_n) \\
 k_{2,x} &= v_n + \frac{h}{2}k_{1,v}, & k_{2,v} &= a\left(x_n + \frac{h}{2}k_{1,x}, v_n + \frac{h}{2}k_{1,v}\right) \\
 k_{3,x} &= v_n + \frac{h}{2}k_{2,v}, & k_{3,v} &= a\left(x_n + \frac{h}{2}k_{2,x}, v_n + \frac{h}{2}k_{2,v}\right) \\
 k_{4,x} &= v_n + hk_{3,v}, & k_{4,v} &= a(x_n + hk_{3,x}, v_n + hk_{3,v})
 \end{aligned}$$

Finally, the simulation computes the true state for the next frame by taking a weighted average of these four slopes:

$$x_{n+1} = x_n + \frac{h}{6}(k_{1,x} + 2k_{2,x} + 2k_{3,x} + k_{4,x}) \quad (4)$$

This creates an exceptionally stable, physically accurate, and determinist real-time collision model.